MYSX-256/512: A 256-bit Block Cipher with 512-bit Keys (v0.1-draft2)

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Abstract

We present MYSX-256/512, a 256-bit block cipher with 512-bit keys and 128 rounds. The design combines Feistel cells with Simon's round function, a Myszkowski byte transposition, word rotations, and an ARX key schedule. We provide the full specification, test vectors, and a self-analysis covering differential, linear, integral, and related-key cryptanalysis. The differential analysis (§3) reports upper bounds on the best single-trail weight over all 31 rotation offsets. Probability-1 differential weight-0 chains exist at some offsets; the longest, 13 rounds (rounds 10–22 mod 31), is the maximum possible (verified by exact linear algebra over the 2^{128} left-half differences, and empirically with 100% propagation over 262,144 pairs across 8 keys). The best single-trail weight found at R=14 ranges from 2 (r0=10) to about 120 across offsets; the 128-bit threshold is reached at R≈24 for r0=10 versus r≈15 for r0=0. The r=4 trail at r0=0 has been directly verified with a clustering factor of 2.00× (observed 2^{-7} vs MILP 2^{-8}). The linear analysis is built on an exact model of F: the LAT admits a closed form (`|corr(b,g)| = 2^{-h(b)}`, `h(b) = rank(A_b)/2` on a 32-cycle), mask propagation is bit-exact, and every trail carries an exact key-dependent sign. Within this model we verify `|corr|`=1 chains over up to 9 consecutive rounds (three independent keys), and exact linear hulls at r=10,11,12 match the real cipher within sampling error ($|z| < 0.4$). A beam search over the exact model, run in both directions (forward and backward) for all 31 rotation offsets and up to 128 rounds, and merged (§4.10), gives achievable single-trail weights of 2 at 12 rounds, 5 at 14 rounds, 38 at 20 rounds, 786 at 64 rounds and 2248 at 128 rounds; after round ~20 the best weight grows by roughly 12–21 per round up to R=112 (the last step, 112→128, is ≈36 per round because only one offset fits a full 128-round trail), and the largest round count with an achievable single trail of weight < 64 (≈2^{128} texts) is 22 (weight 63, a tie with the threshold). Among the tabulated round counts, backward trails are the cheapest at R=9–14, 18, 20, 24–48 and 96–128, and forward trails at R=16, 22, 56, 64 and 80. A hull check against the real cipher shows that the offset-20 backward trail is isolated in its hull at R=14 (weight 5, rounds 100–113, empirical correlation matching 2^{-5}), and an exact budgeted hull (§4.10.5) extends it to R=16 (rounds 98–113): two trails of weight 12 add up to $|\text{corr}| = 2^{-11}$, which a 2^{28}-sample measurement on one key confirms within 1.9σ. This is a verified linear distinguisher on a 16-round window of the cipher (seven weighted rounds followed by a 9-round weight-0 chain), not an attack on the full cipher. At R=17 the best end pairs of offsets 20 backward and 23 forward cancel exactly within the enumerated budget. Other trails can be equal to their single-trail value, a factor ~2 larger, or exactly cancelled (§4.10.3, §4.10.5). No lower bound over 128 rounds is established; the weight-0 windows are a property of 5–9 round windows, not an attack on the full cipher. An integral analysis with standard

division-property rules, extended to all 31 rotation offsets (§4.9), finds integral distinguishers covering 6 rounds with a 2^{32}-text integral (word 0 active) and 5 rounds with a 2^{64}-text integral (words 0 and 4 active) at r0=0. The offsets r0=12 and r0=27, which maximize the weight-0 linear chains, are not special for DP. At r0=15 (shift 16) the distinguishers extend to 10 rounds for both patterns, with a thin tail (one balanced bit at R=10) and practical strength in the 7–8 round range. The R=7/R=6 bits were verified as consistent with a 2-text (or 4-text) toggle; the full 2^{32}/2^{64} integrals were not verified. A 10-round single-bit dependency propagation is also reported but is not an integral distinguisher. The three analysis directions peak at different rotation offsets (differential r0=10, linear free chains r0=12/27, linear hull at offset 20 backward = r0≈5–6 in first-round convention, integral r0=15), so no single-offset bound describes the cipher as a whole. The key schedule was audited with 22 structural and empirical tests up to weight-6 with no persistent Δrk=0 pattern found. The CTR keystream also passes the full NIST SP 800-22 battery (100 sequences × 10^6 bits, fresh key and nonce per sequence, no failures; §4.11), which rules out gross statistical biases but is not evidence of cryptographic security. MYSX has not been externally audited. It is released as v0.1-draft2 for public review and must not be used to protect real data.

1. Introduction

MYSX-256/512 is an experimental block cipher. It is offered as a public proposal for cryptanalysis, not as a production primitive. The author makes no claim of security beyond the observed results reported here, and explicitly welcomes external review.

1.1. Security objectives (and non-objectives)

Model. MYSX is analyzed in the single-key, chosen-plaintext model for confidentiality, combined with an encrypt-then-MAC AEAD wrapper for integrity (Section 2.9). The design does not claim security under related-key, chosen-ciphertext, or side-channel models; see Section 5 for related-key coverage and Section 9 for engineering notes.

Claimed security level. None formally. The paper reports observed upper bounds on single-trail weights (Section 3), a verified exact linear model up to a small number of rounds (Section 4), a full-length beam search over that model in both directions (§4.10), and integral distinguishers at 6 rounds for multi-bit inputs at r0=0, extended to 10 rounds at r0=15 (§4.9). These are inputs to cryptanalysis, not proofs.

Reading thresholds. A differential trail of weight `w` needs about `2^w` chosen pairs; a linear trail of weight `w` has `|corr| = 2^{-w}` and needs about `2^{2w}` known texts. Therefore a 2^{128}-text budget corresponds to a differential weight threshold of ≈128 but to a *linear* weight threshold of ≈64. Differential and linear weights in this paper must not be compared directly; compare `w_diff` with `2·w_lin`.

Non-objectives. MYSX does not claim:

- post-quantum security beyond the trivial Grover bound (2^{256} quantum iterations for a *uniformly random* 512-bit key; keys produced by the HKDF-based API of Section 2.10 from a 256-bit secret give 2^{128}, see Section 6);
- resistance to related-key attacks beyond the heuristic coverage in Section 5;

- resistance to integral attacks beyond the 6–10 round distinguishers reported in §4.9 (offset-dependent; 6 rounds at r0=0, 10 rounds at r0=15), nor resistance to impossible-differential, boomerang, or algebraic attacks (none analyzed);
- resistance to linear-hull effects beyond the sampled checks of §4.6 and §4.10.3–§4.10.5 (no hull-aware bound beyond R=17);
- constant-time execution in JavaScript (side channels out of scope).

1.2. Design rationale

MYSX combines four building blocks:

- **Simon's F** `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`. Chosen for its **studied** diffusion/linear/differential properties in Simon and its simplicity (no S-boxes, only AND/XOR/rotl). It does **not** inherit Simon's security proofs; the surrounding structure is different. See Section 7 for what transfers and what does not.
- **Feistel cells over 32-bit word pairs.** Four parallel cells per round, then a byte transposition over the full 32-byte state, then word rotations. The result is classified as an SPN-like structure with Feistel cells, not as a "2-branch Feistel". See Section 2.1.
- **Myszkowski transposition** (Section 2.3.1). A fixed byte permutation derived from the keyword `MYSZKOWS`. The choice is didactic; it provides byte-level diffusion but no cryptographic argument beyond that. An optimized permutation would provide strictly better diffusion per round; see Section 9.
- **ARX key schedule** (Section 2.4). A bijective Feistel 8+8 update with modular addition, XOR, and rotations. Chosen for simplicity and to avoid S-boxes; analyzed in Section 5.

The parameter choices (256-bit block, 512-bit key, 128 rounds) are not derived from a formal argument. They were chosen to:

- provide a large block (avoid birthday-bound issues of 64-bit blocks);
- provide a large key (Grover leaves 256-bit security for a random 512-bit key);
- provide a large round count relative to Simon128/128 (68 rounds) as a precaution against unknown trails.

No claim is made that 128 rounds is sufficient. The round count is a parameter, not a security level.

1.3. Comparison with related designs

Design	Block	Key	Rounds	Structure	S-boxes
Simon128/128	128	128	68	Feistel	No
Speck128/256	128	256	34	ARX	No
ChaCha20	512	256	20	ARX	No
MYSX-256/512	256	512	128	SPN-like with Feistel cells	No

MYSX's closest sibling is Simon. The differences (block size, branch organization,

transposition, word rotations, non-linear key schedule) are substantial enough that Simon cryptanalysis does not transfer directly (see Section 7).

Performance. MYSX is estimated to run at a few MB/s in JavaScript, versus hundreds of MB/s for AES-GCM via WebCrypto (AES-NI). MYSX is not competitive on performance; it is a research primitive.

1.4. Versioning and revision notes

Versioning. The reference implementation (`mysx.prod.js`) is labelled `v0.1-draft` and has not changed since. This document is the second documentation draft of that same design and is therefore labelled `v0.1-draft2`. An earlier revision of this text carried the label `v0.2-draft`; that label is withdrawn because it suggested a change of the cipher or of the code, and there is none.

Changes since the previous draft (`v0.1-draft`).

- **New direction (linear):** the beam search (§4.10) now supports `--direction backward` (the offset is the index of the *last* round of the trail, mod 31) and a `merge` command that combines results from both directions. The table of §4.10.2 is now the minimum over forward and backward and over all offsets; a few `v0.1-draft` checkpoint values (R=12, R=48, R=112, R=128, and in general those where backward wins) are superseded.
- **New verified linear distinguisher (on a 16-round window):** the offset-20 backward trail is isolated in its hull at R=14 (weight 5, rounds 100–113, `hull_count` = 1 trail, empirical correlation 2^{-5} on the real cipher, §4.10.3) and, by the exact budgeted hull of §4.10.5, extends to R=16 (rounds 98–113, |corr| = 2^{-11}). This is the first non-trivial linear distinguisher the analysis certifies. It consists of seven weighted rounds (98–104, total weight 12) followed by a 9-round weight-0 chain (105–113).
- **New exact hull at R=15–17 (§4.10.5):** `hull_bidir.py`, `scan_endpairs.py`, `test_signed_dp.py`, `emp_check.py`. The R=16 row of §4.10.2 (weight 8, offset 23 forward) cancels; the best certified correlation at R=16 is 2^{-11}.
- **Correction on R=128:** in forward mode only r0=0 fits a 128-round trail. In backward mode, offset 3 gives `end_round(3) = 127`, `realizable_R = 128`, so the combined R=128 = 2248 at offset 3 backward **is a real trail inside the cipher**, not a virtual extension. The `v0.1-draft` statement that "for r0>0 the value at R=128 is a virtual extension" is forward-specific; §4.10.2 clarifies.
- **Differential offset sweep (§3.5, new).** The rotation offset was varied over all 31 values for the first time. Probability-1 differential weight-0 chains exist at some offsets; the longest, 13 rounds (rounds 10–22 mod 31), is the maximum possible, verified by exact linear algebra over the 2^{128} left-half differences and empirically with 100% propagation over 262,144 pairs × 8 keys. At R=14 the best single-trail weight found ranges from 2 (r0=10) to about 120 across offsets; the 128-bit threshold is at R≈24 for r0=10 versus r≈15 for r0=0. The differential bounds of §3.2, which were measured at r0=0 only, are **not** representative of all offsets.
- **Integral offset sweep (§4.9).** All 31 offsets were tested at the saturation rounds of r0=0. The offsets r0=12 and r0=27 (which maximize the weight-0

linear chains) are **not** special for DP. The peak block is r0 ∈ [12,17], with a maximum at r0=15 (shift 16). At r0=15 the distinguishers extend to 10 rounds for both the word0 and word0,word4 patterns. The R=7/R=6 bits were verified as consistent with a 2-text (or 4-text) toggle; the full 2^{32}/2^{64} integrals were not verified.

- **Three geographies.** The three analysis directions peak at different rotation offsets: differential at r0=10, linear free chains at r0=12/27, linear hull at offset 20 backward (≈r0=5–6 in first-round convention), integral at r0=15. None of the single-offset bounds previously reported describes the cipher as a whole.
- **Hull check (§4.10.3)** extended: the r0=23 forward trail continues to cancel from R=8; the r0=24 forward trail cancels from R=8; the r0=12 forward trail is exact up to R=12; the offset-20 backward trail is exact from R=1 to R=14. Hull effects are not uniform.
- **Multi-bit starts:** explored (`pairs47`, `rand:5000:4`) and do not improve on single-bit starts at any R. `pairs47` is a small structured subset; `rand:5000:4` covers 5000 masks out of 2^{256} and is not conclusive for structured multi-bit masks. The honest statement is "the multi-bit starts tested do not improve on single-bit".
- Added §4.10.1 backward-step rule and its `selftest` (every forward edge is a backward edge with the same cost, 325 edges checked).
- Abstract: numbers updated to the combined forward+backward values; added the verified distinguisher at R=14.

Corrections made in `v0.1-draft2` (errors found while reviewing the previous text; no experimental result changed):

- §3.3: the direction of the "systematic model error" argument was inverted. An observed probability *higher* than predicted means the model *over*estimates the cost (true weight 7, not 9). Fixed.
- §3.2: the r=4 trail is not a row of the table; the text now says so.
- §3.1/§3.2/§3.4 now state that the rotation offset was not varied in the differential runs of `v0.1-draft`, which makes the r≈15 extrapolation optimistic; §3.5 replaces that limitation with the offset sweep.
- §4.10.2 and the abstract: the statement on which direction is cheapest contradicted the table; rewritten from the table. The "15–25 per round" growth figure did not match the table (12.5–20.9 per round up to R=112, ≈36 for 112→128); replaced.
- §4.10.3, abstract, §4.7, §10: "distinguisher at 14 rounds" now says "on a 14-round window (rounds 100–113)" and explains that the weight is flat from R=5 to R=14 (§4.10.5 later extends the window to 16 rounds).
- §4.9, §7: Simon integral figures and the Simon literature list corrected against the sources (Simon32: 14 rounds with conventional bit-based division property, 15 with three subsets; the Simon128 26-round integral is Xiang et al.; ePrint 2014/973 is Shi et al.; the 49-round Simon128/128 result is a key-recovery attack by Chen–Wang; Biryukov–Roy–Velichkov is FSE 2014).
- §4.10.4: the hull gain needed at R=24 (w=88) to reach the 2^{128} budget is 2^{24} in correlation (w − 64), not 2^{40}.
- §1.1, §6: the Grover 2^{256} figure holds for a uniformly random 512-bit key only; HKDF-derived keys give 2^{128}. The HKDF sentence in §6 was imprecise and is fixed.

- §8: functional test count aligned with `README.md` and `SECURITY.md` (17 tests).
- §9.2: note that 128-bit random nonces make accidental reuse negligible, so `NonceGuard` mainly enforces the per-key message limit.
- **Differential claims corrected (§3.1, §3.2, §3.4, §3.5).** The differential runs of `v0.1-draft` were measured at r0=0 only. The offset sweep of §3.5 shows that r0=0 is near the top of the distribution (115 in the beam at R=14, versus a maximum of ~121 at r0=1), so the §3.2 numbers are valid upper bounds for r0=0 but are not representative. The time-limited MILP values reported for r0≠0 during the offset sweep are upper bounds with `gapRel=0.2`, not minima. The "three trails" of the earlier draft of this text were three points on the same 13-round chain, not independent chains; the independent deterministic chains are two (the 13-round chain at r0=10 and the 6-round chain at r0=15).
- **Integral claims clarified (§4.9).** The R=7/R=6 balanced bits at r0=15 are verified as consistent with a 2-text or 4-text toggle, not as full $2^{32}/2^{64}$ integrals; the tail at R=8–10 is a model prediction. The "shift 16 is structurally the weakest" reading is a hypothesis, not established.

2. Specification

2.1. Parameters

Parameter	Value
Block size	256 bits (8 words of 32 bits)
Key size	512 bits (16 words of 32 bits)
Rounds	128
Structure	SPN-like: 4 parallel Feistel cells + byte transposition
Non-linearity	`F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`
Diffusion	Myszkowski transposition (keyword `MYSZKOWS`)
Rotation	`rotl(word, (r mod 31) + 1)` per word and round
Final whitening	XOR with 8 words from the key schedule

Structural note. MYSX is sometimes described as a "2-branch Feistel" by analogy with Simon. Strictly, each round applies four independent Feistel cells over 32-bit word pairs, then a byte transposition over the full 32-byte state, then word rotations. It is more accurately classified as an SPN with Feistel cells. The "Feistel" label is a convenience.

2.2. Notation

- `rotl(x, n)`: circular left rotation of `x` by `n` bits.
- `∧`: bitwise AND.
- `⊕`: bitwise XOR.
- `s[r]`: state at round `r` (`s[0]` = plaintext, `s[128]` = ciphertext).
- `s[r][i]`: word `i` (0..7) of the state at round `r`.
- `rk[r][i]`: subkey `i` (0..7) of round `r`.
- `rkFinal[i]`: word `i` (0..7) of the final whitening.

2.3. Constants

```
CONST[0..7] = [
  0x243f6a88, 0x85a308d3, 0x13198a2e, 0x03707344,
  0xa4093822, 0x299f31d0, 0x082efa98, 0xec4e6c89
]
```

These are the fractional digits of π (same as Blowfish).

2.3.1. Myszkowski permutation

Keyword: `MYSZKOWS` (8 letters, S repeated at positions 3 and 8).

Alphabetical order: K(5), M(1), O(6), S(3,8), W(7), Y(2), Z(4).

The byte permutation `MYZ_PERM[0..31]` is constructed as:

```
MYZ_PERM = [
   4,12,20,28,                 # K (column 4)
   0, 8,16,24,                 # M (column 0)
   5,13,21,29,                 # O (column 5)
   2, 7,10,15,18,23,26,31,     # S,S (columns 2 and 7, read row-wise)
   6,14,22,30,                 # W (column 6)
   1, 9,17,25,                 # Y (column 1)
   3,11,19,27                  # Z (column 3)
]
```

The inverse permutation `MYZ_INV` satisfies `MYZ_INV[MYZ_PERM[i]] = i`.

Design note. The Myszkowski choice is didactic (classical transposition with a keyword) and adds no cryptographic argument. A byte permutation optimized for diffusion (e.g., a structured permutation over the 4×8 byte grid) would strictly dominate it. We keep Myszkowski in this draft because it is the design under review; a future version may replace it.

2.4. Key schedule

2.4.1. Initialization

Input: key `K[0..63]` (64 bytes).

Initial state `s[0..15]` (16 words of 32 bits):

```
s[i] = big_endian_word(K[4i .. 4i+3])
```

2.4.2. Expansion

For `r = 0 .. 127`:

```
spread = (r + 1) * 0x9E3779B9  (mod 2^32)

# Round subkeys
for i in 0..7:
```

```
    rk[r][i] = s[i] ⊕ s[(i+5) mod 16] ⊕ s[(i+11) mod 16]
               ⊕ CONST[(r+i) mod 8] ⊕ spread

# For i = 0..3: fold fk into rk
for i in 0..3:
    fk = s[(i+8) mod 16] ⊕ CONST[(r+i+3) mod 8]
    rk[r][i] = rk[r][i] ⊕ fk

# State update (Feistel 8+8, bijective)
L = s[0..7]
R = s[8..15]
for i in 0..7:
    a = R[i]
    b = R[(i+1) mod 8]
    c = R[(i+3) mod 8]
    F = (rotl((a + b) mod 2^32, 7) ⊕ rotl(c, 13) ⊕ CONST[i] ⊕ spread)
    s[i]     = R[i]
    s[i + 8] = L[i] ⊕ F
```

2.4.3. Final whitening

After the 128 rounds:

```
for i in 0..7:
    rkFinal[i] = s[i] ⊕ s[(i+8) mod 16] ⊕ CONST[(128+i) mod 8]
```

2.5. Round function

Each round `r` transforms state `s[r]` into `s[r+1]` in four steps:

Step 1. AddRoundKey.

```
for i in 0..7:
    s[r][i] = s[r][i] ⊕ rk[r][i]
```

Step 2. Feistel cells. For `p = 0..3`:

```
L = s[r][p]
R = s[r][p + 4]
s_mid[p]     = R
s_mid[p + 4] = L ⊕ F(R)
```

where `F(x) = (rotl(x,1) ∧ rotl(x,8)) ⊕ rotl(x,2)`.

Step 3. Myszkowski transposition. Convert `s_mid[0..7]` to bytes (big-endian per word):

```
bytes[4i .. 4i+3] = big_endian(s_mid[i])
permuted[j] = bytes[MYZ_PERM[j]]  for j = 0..31
s_perm[i] = big_endian_word(permuted[4i .. 4i+3])
```

Step 4. Word rotation.

```
shift = (r mod 31) + 1
for i in 0..7:
    s[r+1][i] = rotl(s_perm[i], shift)
```

2.6. Encryption

```
s[0] = plaintext  (as 8 big-endian words)
for r = 0..127:
    apply round r (Section 2.5)
for i in 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
ciphertext = s[128]  (as 32 big-endian bytes)
```

2.7. Decryption

Inverse of encryption:

```
s[128] = ciphertext
for i in 0..7:
    s[128][i] = s[128][i] ⊕ rkFinal[i]
for r = 127..0:
    apply inverse round r (inverse of Section 2.5, reverse order)
plaintext = s[0]
```

The inverse round consists of:

1. Inverse rotation: `rotl(word, 32 - shift)`.
2. Inverse transposition: `bytes[MYZ_PERM[j]] = permuted[j]`.
3. Inverse Feistel: `R = newL`, `L = newR ⊕ F(R)`.
4. XOR with `rk[r][i]`.

2.8. Mode of operation (CTR)

32-byte counter:

```
counter[0..15]  = nonce (16 bytes)
counter[16..31] = 0  (128 bits, big-endian)
```

For each 32-byte plaintext block:

```
keystream = MYSX_encrypt(counter)
ciphertext_block = plaintext_block ⊕ keystream
counter[16..31] += 1  (big-endian)
```

2.9. AEAD (encrypt-then-MAC)

Packet: `VERSION(1) ‖ NONCE(16) ‖ CT(n) ‖ TAG(32)`

MAC input: `VERSION ‖ NONCE ‖ AAD_LEN(4 BE) ‖ AAD ‖ CT_LEN(4 BE) ‖ CT`

TAG: `HMAC-SHA256(macKey, MAC input)`

Verification: the decryptor verifies the TAG before decrypting.

2.10. Key derivation

From a random secret (HKDF-SHA256).

```
encKey = HKDF-SHA256(secret, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(secret, salt, info = context ‖ ":mac", L = 32)
```

From a password (PBKDF2-SHA256 + HKDF).

```
prk    = PBKDF2-SHA256(password, salt, iterations = 600000, L = 32)
encKey = HKDF-SHA256(prk, salt, info = context ‖ ":enc", L = 64)
macKey = HKDF-SHA256(prk, salt, info = context ‖ ":mac", L = 32)
```

2.11. Test vectors

See `KAT.json` (public). Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**.

3. Differential cryptanalysis

3.1. Model

Tool: `misc/mysx_milp2.py` (PuLP + CBC/HiGHS, with `--r0` rotation offset). AND model: `a = Δx ∨ Δy`, which counts a cost-1 event whenever at least one input is active. For an AND with independent inputs, this is the exact per-bit cost. The real approximations in the model are:

- The Markov-independence assumption does not hold: the two rotated inputs of F come from the same word `x`.
- The reported runs use a single input difference per offset (the one predicted by the free chain table of §3.5 as maximizing the free chain, or w2b30 for the §3.2 table); multi-bit input differences may yield lower weights.
- `gapRel=0.2` and `timeLimit=300–900` return feasible solutions (upper bounds), not certified optima. The MILP is therefore used as a lower-cost exploration tool; the **beam search** of §3.5 is the primary source for the offset-by-offset weight table.
- The model captures single trails, not differentials (clusters of trails with the same output difference).

The offsets were varied for the first time in this draft (§3.5). The §3.2 table, measured at r0=0 only, is a valid upper bound for r0=0 but is **not** representative of all 31 rotation offsets.

3.2. Observed upper bounds (r0=0 and the offset sweep)

At r0=0. The §3.2 table of `v0.1-draft` remains valid as an upper bound for that offset:

Rounds	Input	Weight (upper bound)	Single-trail prob.
5	w1b24	2	2^{-2}
6	w1b24	6	2^{-6}
7	w2b30	10	2^{-10}
8	w2b30	18	2^{-18}
12	w2b30	68	2^{-68}
14	w2b30	114	2^{-114}

Extrapolation at r0=0: the 128-bit threshold is reached at r≈15. **This is an extrapolation, not a proof.** It tells us what the model produces for that input at that offset.

Offset sweep (§3.5). At R=14, the best single-trail weight found by the beam search (multi-input, forward only) ranges from **2 at r0=10** to about **120** across offsets. A representative sample (beam, forward, single-bit starts):

r0	0	1	10	22	29	30
best weight at R=14	115	121	**2**	107	119	118

The numbers for r0≠10 are upper bounds from a beam, not proven minima; the maximum observed is 121 (r0=1), not 114 (r0=0). **The r0=0 extrapolation of r≈15 is therefore one point in a distribution that runs from r≈24 (r0=10) to r≈15 (r0=0) and above.** The `v0.1-draft` statement that "the extrapolation to r≈15 is likely optimistic" is superseded: the r≈15 value is correct for r0=0 but is not the cipher-wide threshold.

Growth at r0=10 (input w1b31, beam, multi-input): 2, 4, 6, 10, 15, 22, 34, 49, 78, 100, 133 at R=14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24. Linear interpolation puts the 128-bit threshold at **R≈23.5**. A direct MILP run at R=24 gave 195, which is a looser upper bound (`gapRel=0.2`, time limit); the beam value of 133 is tighter. The beam is the primary source for this table.

Comparison with the linear direction. At 14 rounds the differential upper bound at r0=0 is 114 ($\approx 2^{114}$ pairs); at r0=10 it is 2 ($\approx 2^{2}$ pairs, the deterministic chain of §3.5). The linear side has a **verified distinguisher on a 14-round window** (offset-20 backward trail, rounds 100–113, weight 5, $\approx 2^{10}$ texts, isolated hull; §4.10.3). The differential direction has a deterministic 13-round chain (r0=10, §3.5), which in data cost (1 pair) is stronger than the linear distinguisher, but is 3 rounds shorter. Neither direction is uniformly dominant; both peak at different offsets. The v0.1-draft conclusion that "the linear direction is the weaker one by a wide margin" is superseded: the gap noted there was not intrinsic (the linear figure is the minimum over all 31 rotation offsets, while the differential figure was for one input at one position).

Direct verification at r=4 (w=32). The r=4 trail (it is not a row of the table above: Δin = {w0b0}, MILP weight 8, Δout = {w0b4, w1b10, w1b16, w1b30, w6b4}) was verified with 8 fixed keys × 2^{17} pairs each, using the full 128-round cipher with the final whitening disabled (to isolate the round function). Observed hit rate: 7.802×10^{-3} over 1,048,576 pairs. Expected from MILP: $2^{-8} = 3.906 \times 10^{-3}$. **Ratio: 2.00×.**

The trail exists at r=4. The single-trail probability is a factor 2 higher than the MILP predicts, consistent with the clustering factor documented in §3.3. **The MILP model is correct within a factor of 2 at this round count.** This measurement was made at r0=0; the clustering factor at other offsets has not been measured (§3.4).

3.3. Empirical check (r=4, w0b0)

As described above, the direct verification gives a ratio of 2.00× between the observed probability and the MILP single-trail prediction. Two interpretations are consistent with the data:

- **Differential clustering** (more likely): multiple trails with the same output difference Δout contribute to the observed probability, so the effective probability is higher than the single-trail probability.
- **Model systematic overestimate of the cost** (less likely): the AND model `a = Δx ∨ Δy` charges one bit too much at r=4. This would imply the true weight is 7, not 8. (Because the observed probability is *higher* than predicted, the model cannot be *under*estimating the cost; an underestimate would give a probability lower than predicted and a true weight of 9.)

Without a full clustering analysis (summing over all trails with the same Δin, Δout), we cannot distinguish these two. The practical implication is the same: the effective weight of the r=4 trail at r0=0 is 7, not 8, so the MILP weight is the pessimistic side by one bit.

What this does to the extrapolation. A *constant* factor 2 is one bit of weight; at the observed 8–23 bits of weight per round it moves the "128-bit threshold at r≈15" by a small fraction of a round, not by 2–3 rounds. Only a factor that grows with the weight (for example the ratio 7/8 applied to the whole trail) would matter: it would move the crossing to roughly r≈16–17. The measurement was made at r0=0; at r0=10, where the deterministic chain exists, the clustering factor is not characterized (§3.4). The extrapolation should be read as approximate and, see §3.1, as specific to the input and offset tried.

3.4. What this section does not establish

- A lower bound on the best single-trail weight for any offset. The beam values of §3.5 are upper bounds from a specific start distribution; the MILP values are upper bounds with `gapRel=0.2`.
- An exact clustering analysis (requires modeling all trails with the same Δout), and in particular the clustering factor at r0 ≠ 0.
- Resistance to related-key differentials.
- That the factor-2 clustering is constant across all round counts.

- That the beam, with single-bit starts and 3 candidates per cell, is exhaustive. The exact maximum chain length of 13 rounds for weight-0 chains (§3.5) **is** proven by linear algebra; the non-zero-weight tables are not.

3.5. Differential weight-0 chains (new)

The differential model of §3.2 has the same structure as the linear model of §4.3: a round has weight 0 iff the difference does not touch the right half (words 4..7) of the state, in which case the round acts as a pure bit permutation on the left half. The free chain length depends on the rotation offset.

Free chain table. The maximum number of consecutive weight-0 rounds as a function of the offset r0 (first round mod 31), for a single-bit difference in words 0..3 of the state:

r0	10	11	12	13	14	15	16	17	18	0–6, 27–29
max rounds	**13**	12	11	10	9	8	7	6	5	**3**

The worst offsets are r0 ∈ {0,1,2,3,4,5,6,27,28,29} (3 rounds); the best is r0=10 (13 rounds). The §3.2 table was measured at r0=0, one of the 3-round offsets.

Maximum exact. By exhaustive linear algebra over the 2^{128} left-half differences, no weight-0 differential chain longer than 13 rounds exists for **any** offset, and **not even for multi-bit differences**. The 13-round chain at r0=10 is therefore the exact maximum, not a search artifact.

Verified deterministic chains. Two independent chains were verified against the real cipher with fixed keys, 2^{15} pairs per key, 8 keys each (262,144 pairs total), full 128-round cipher, final whitening disabled:

Offset	Input	Output	Rounds	Propagation
r0=10	w1b31	w4b12	**13**	**100.0000%**
r0=15	w2b30	w4b5	6	100.0000%

The 13-round chain at r0=10 covers rounds 10–22, 41–53, 72–84 and 103–115 of the cipher (period 31). It is the same chain at every occurrence.

The chains of 11 and 8 rounds reported during the sweep are **not** independent: they are the same 13-round chain started at a later point (w1b22 is the mask at round 12 of the 13-round chain, w2b24 is the mask at round 15). Reporting them as “three verified chains” is circular; the independent chains are the 13-round chain and the 6-round chain.

Growth at r0=10 (input w1b31, beam, multi-input, forward): 2, 4, 6, 10, 15, 22, 34, 49, 78, 100, 133 at R=14..24. The R=16 (weight 6) and R=17 (weight 10) trails were verified empirically by the external reviewer: observed 1.63×10^{-2} against a prediction of 1.56×10^{-2} at R=16, and 1.1×10^{-3} against 9.8×10^{-4} at R=17, within sampling error. The 128-bit threshold is at R≈23.5.

Three geographies. The differential, linear and integral analyses peak at different rotation offsets (first-round convention):

Direction	Peak offset (r0)	Longest verified	Data cost
Differential	**r0=10**	13 rounds (weight 0)	**1 pair**
Linear free chain	r0=12, r0=27	9 rounds (weight 0)	2^0
Linear hull	offset 20 backward ≈ r0=5–6	16 rounds (hull)	2^{22} texts
Integral	**r0=15**	10 rounds	2^{32} / 2^{64} texts

No single offset dominates all three. This is the main structural finding of the offset sweep: **the three analysis directions must be reported offset by offset, not by a single representative offset.**

4. Linear cryptanalysis

4.1. Overview

An earlier version of this paper (§4 of the v0.0 draft) reported only negative results: three approximate linear MILP models (`misc/mysx_linear_milp.py`) all produced trails that failed empirical verification at w=32. That negative result has now been explained. The models used per-cell AND activity (`act = α ∨ β ∨ γ`) as a proxy for correlation, which is not the correct structure of F's LAT. A correct model exists in closed form and is implemented in `cryptanalysis/models/linear/mysx_linear_model.py` (with detailed documentation in `cryptanalysis/models/linear/LINEAR_MODEL.md` and raw data in `cryptanalysis/models/linear/linear_model_data.json`). The model is machine-checked against the real cipher by `python3 mysx_linear_model.py selftest`.

Result in one line. The LAT of F is exactly `corr(b,g) = ±2^{−h(b)}`, where `h(b)` is half the GF(2)-rank of a 32-cycle subgraph determined by `b`; there is a free choice of `g` among `4^{h(b)}` values with sign determined by a quadratic form. Mask propagation is bit-exact; every trail carries an exact key-dependent sign. Verified consequences: `|corr|=1` chains over up to 9 consecutive rounds (three keys), and exact linear-hull predictions at r=10,11,12 that match the real cipher within sampling error ($|z| < 0.4$). §4.10 extends the search for cheap trails to 128 rounds, all rotation offsets, and **both directions** (forward and backward), and shows that single-trail weights must be read together with hull effects. For the offset-20 backward trail at R=14 (weight 5, covering rounds 100–113), the hull check is complete: the trail is isolated and its empirical correlation matches 2^{-5} (§4.10.3). An exact budgeted hull (§4.10.5) extends the verified window to R=16 (rounds 98–113, $|\text{corr}| = 2^{-11}$). This is a verified linear distinguisher on a 16-round window of the cipher.

4.2. Exact LAT of F

For output mask `b` and input mask `g`,

```
b.F(x) ⊕ g.x  =  Q_b(x) ⊕ l.x ,   l = rotr(b,2) ⊕ g ,
Q_b(x) = ⊕_j b_{j+1} x_j x_{j−7} .
```

`Q_b` is a quadratic form whose graph `G_b` is a subgraph of the 32-cycle `j — j−7` (edge `(j, j−7)` present iff `b_{j+1}=1`; a single cycle because `gcd(7,32)=1`). Let `A_b` be the GF(2) adjacency matrix of `G_b`. Then:

- `corr(b,g) ≠ 0` iff `l = A_b y` for some `y`. There are exactly `2^{rank(A_b)} = 4^{h(b)}` valid `g`.
- For each valid `g`, `corr(b,g) = (−1)^{Q_b(y)} · c_0(b)` with `|c_0(b)| = 2^{−h(b)}`, where `h(b) = rank(A_b)/2`.
- `h(b)` equals the maximum matching = minimum vertex cover of `G_b` (König). The only graph that is not a forest is `b = 0xFFFFFFFF` (full cycle), where `rank = 30` and `h = 15`.
- Parseval holds: `Σ_g corr(b,g)² = 1`.

This is **exact** and **machine-checked** against brute-force LAT of F for random `b` of Hamming weight up to 8 (`selftest`).

Cost data.

- Single-bit `b`: `h = 1`, four valid `g`, `|corr| = 1/2`.
- Two-bit `b` (496 masks): histogram `{2: 464, 1: 32}` (the 32 pairs at bit distance 7 are adjacent edges).
- Uniform random 32-bit `b` (16384 samples): histogram of `h` peaks at `h = 11` (4130 samples); full histogram in `linear_model_data.json`.
- Densest masks per unit of weight (bits `(1−7t) mod 32`, `t < 2m`): every extra 2 consecutive edges costs 1; cost per active mask bit approaches 1/2 in the best case, never below 15/32.

4.3. Exact mask propagation (forward and backward)

Forward step. For a round `r`, with input mask `m` on `s[r]` and cells `p = 0..3` (`L = s[r][p]`, `R = s[r][p+4]`):

```
b_p     = m[p]                          (only words 0..3 decide the cost)
g_p   ∈ valid_gammas(b_p)          (4^{h(b_p)} choices, each |corr| = 2^{−h(b_p)})
n[p+4]  = m[p] ;  n[p]  = m[p+4] ⊕ g_p
m_next  = lin_fwd(n, r)            (byte transposition, then rotl by (r mod 31)+1)
w_r     = Σ_p h(m[p])
sign    = (Π_p sign_p) · (−1)^{m · rk[r]}
```

Backward step. For a round `r`, given `m_next` on `s[r+1]`:

```
n     = lin_inv(m_next, r)         (inverse rotation, then inverse transposition)
b_p     = n[p+4]                      (this is the output mask of F at cell p)
m[p]    = n[p+4]                          (cost h(n[p+4]) at this round)
m[p+4]  = n[p] ⊕ g_p ,  g_p ∈ valid_gammas(b_p)
```

The `selftest` verifies that every forward edge is a backward edge with the same cost (325 edges checked), so the two searches enumerate the same set of trails. Trails are always stored in forward order (mask on `s[first]` … mask on `s[last+1]`).

Correlation of a trail: `Π_r 2^{−w_r} · (−1)^{Σ_r m_r·rk[r]} · Π_p sign_p`. The linear hull at fixed key is the exact sum over all trails with the same `(m_0, m_R)`; `propagate()` computes it by correlation-vector propagation (pruned by weight). ELP mode (`propagate(..., elp=True)`) gives the key-independent potential under the (incorrect for a deterministic schedule) independent-round-key assumption.

Free-chain structure. A round has weight 0 iff words 0..3 of its input mask are zero; the round then acts as a pure bit permutation and the tracked bit never meets `F`. Since the rotation schedule has period 31, there are only 31 distinct windows. The exact free-chain table (all 128 single-bit masks in words 4..7, 31 windows) is in `LINEAR_MODEL.md` §4 and `linear_model_data.json`. Headline: the longest weight-0 chain is **9 consecutive rounds**, at windows `r ≡ 12 (mod 31)` and `r ≡ 27 (mod 31)`.

4.4. Verified `|corr| = 1` chains

Each of the 9-round chains has been checked against the real cipher (fixed random keys, 2^{21} plaintexts, reduced rounds, final whitening disabled to isolate the round function). Representative results:

window r0	rounds	start bit	end bit	key seed	predicted sign	empirical corr	σ
12	9	s[12].174	s[21].79	1	+1	+1.00000	6.9e-04
12	9	s[12].174	s[21].79	2	−1	−1.00000	6.9e-04
12	9	s[12].174	s[21].79	3	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	1	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	2	+1	+1.00000	6.9e-04
27	9	s[27].146	s[36].55	3	+1	+1.00000	6.9e-04
0	5	s[0].208	s[5].55	1	−1	−1.00000	6.9e-04
0	5	s[0].208	s[5].55	2	−1	−1.00000	6.9e-04
0	5	s[0].208	s[5].55	3	−1	−1.00000	6.9e-04

The magnitude is 1 for every key. Only the sign depends on the key, through `Σ_r m_r · rk[r]`. The model predicts that sign exactly. Per- round masks and complete details are in `linear_model_data.json` (`verified_chains`).

The backward search finds the same 9-round weight-0 chains at the corresponding end residues (20 and 4 mod 31). The `selftest` verifies `|corr|=1` on the real cipher for both, and that the forward and backward chains are the same trails (stored in forward order).

4.5. Best single trails beyond the free chain (upper bounds)

Layered beam search with the exact rules, starts = all 256 single-bit masks, beam 1200, cells with `h ≤ 2` expanded. Each reported weight is an **achievable single-trail weight**, i.e. an upper bound on the best weight, not a minimum.

rounds R	1–9	10	11	12	13	14
best weight, window r0=12	0	1	2	3	7	12
best weight, window r0=27	0	1	3	3	6	11

This table covers only the two windows with the longest free chain. §4.10 searches all 31 offsets in both directions and shows that the windows 12 and 27 are the best only up to R=12; from R=13 on, other offsets (notably r0=23 forward and r0=20 backward) give lower single-trail weights.

Note that this is *the opposite* of a linear-distinguisher risk at low rounds: at the 9-round mark we still have weight 0 (correlation ±1 in absolute value). The reader should not confuse this with an attack on the full cipher — see §4.7.

4.6. Hull verification (r0=12, exact signed hulls)

The single-trail numbers above are inputs to a hull computation. For three trails at r0 = 12 with weights 1, 2, 3, the model produces an exact signed prediction (fixed key, sum over all trails of weight ≤ w+4 with the same end masks) and the real cipher produces an empirical correlation over 2^{22} plaintexts.

window r0	R	best-trail w	hull masks kept	predicted corr	empirical corr	σ	z
12	10	1	4	+0.500000	+0.499997	4.9e-04	−0.01
12	11	2	40	−0.250000	−0.250008	4.9e-04	−0.02
12	12	3	3152	−0.125000	−0.124813	4.9e-04	+0.38

Three independent predictions, three matches, |z| < 0.4 in every case. The hull sum is not an approximation here; it is the exact polynomial expansion of the trail system in the model, and it is what the real cipher produces within sampling error.

§4.10.3 extends this check to the saved R=14 trails from the beam search. The r0=23 forward trail cancels from R=8; the r0=24 forward trail cancels from R=8; the r0=12 forward trail is exact up to R=12; the offset-20 backward trail is exact from R=1 to R=14. Hull effects are not a uniform correction.

4.7. What this establishes and what it does not

Established (machine-checked by `selftest`):

- Exact LAT of F (valid set, magnitude, sign) against brute force.
- Exact per-round mask rule and key-sign tracking, in both directions (forward and backward edges match, 325 edges checked).
- Existence of key-independent-magnitude `|corr| = 1` single-bit relations over up to 9 consecutive rounds; found by both forward and backward searches at the corresponding residues.

- Exact signed linear-hull predictions (not approximations) at r=10, 11, 12 against the real cipher.
- The `mysx.prod.js` Python port reproduces `KNOWN_KAT_DIGEST`, so all claims are made against the actual reference implementation.
- (§4.10) The beam-search tool reproduces the §4.5 table exactly, its per-round masks agree with the real cipher wherever a trail is its own hull, and the `selftest` runs in both directions.
- (§4.10.3, §4.10.5) The offset-20 backward trail at R=14 (weight 5, rounds 100–113) is isolated in its hull (`hull_count = 1` trail) and its empirical correlation matches 2^{-5} on the real cipher; the exact budgeted hull extends it to R=16 (rounds 98–113, $|\text{corr}| = 2^{-11}$, confirmed within 1.9σ on one key). This is a **verified linear distinguisher on a 16-round window**. It is seven weighted rounds followed by the 9-round weight-0 chain of this section, so it is the free chain plus a short weighted prefix, not new evidence about long trails.

Not established:

- A **lower bound** on the best trail weight over 128 rounds. Beam search gives upper bounds from a specific start distribution.
- An exact 128-round hull. Even the pruned `propagate()` will not scale; a certified bound would need a SAT/MILP encoding.
- A hull-aware correlation for the trails found in §4.10 beyond R=14. §4.10.3 shows that single-trail weight can over- or under-estimate the real correlation (cancellation was observed for some trails and isolation for others).
- The cost of a key-recovery attack.
- Behaviour of multi-bit starts at larger R beyond the pairs47 and rand:5000:4 runs (which did not improve on single-bit; rand:5000:4 is too small to be conclusive for structured multi-bit masks).
- Anything about the last-round / whitening structure, related-key linear behaviour, or side channels.

The weight-0 chains are not an attack. A 9-round `|corr| = 1` relation sits in a 128-round cipher. Standard linear cryptanalysis extrapolates a weight over the full length; here the trail weight over rounds 0..8 is 0 and the weight over rounds 9..127 is unknown. The result is best read as: *the exact machinery is in place and it agrees with the real cipher on small round counts*, not as *MYSX is broken*. The 128-round picture remains open, although §4.10 shows that the cheap region ends at roughly 20 rounds in the single-trail model.

4.8. Previous incorrect models and why they failed

For the record, the v0.0 draft reported three approximate models, all with negative empirical verification. They used per-cell AND activity as a proxy for correlation, which is not the correct structure of F. The correct structure is the rank of the graph `A_b` above. The previous models produced trails with predicted `|corr| ≈ 2^{-8}` at r=4 but no detectable signal — consistent with the fact that those trails have zero true correlation in the exact model.

An instructive detail from the old work remains relevant to *any* future linear work on MYSX: the empirical verification harness must use **fixed keys** per sample.

With random keys per plaintext, the sign of every linear trail is key-dependent and the observed correlation of a genuinely nonzero trail collapses to zero. See `misc/validate_lineal_harness.js`, which validates the harness against a hand-computed `w=8` case (observed correlation 0.4993 vs 0.5).

4.9. Integral / division property (preliminary, limited)

Tool: `misc/mysx_division_property_v2.py` with standard division- property rules (COPY: `k_a = Σ k_copies`; XOR: `k_c = k_a + k_b` with `(1,1)` invalid; AND: `k_c = k_a OR k_b`). The XOR and OR variants were found to be equivalent for all input patterns tested (the `(1,1)` constraint never fired), so the reported results do not depend on the choice between them.

Results at r0=0.

Input	Active bits	Integral size	Rounds with balanced bits	Saturation
bit 0 of word 0	1	2	10	R = 11
word 0	32	2^{32}	6	R = 7
words 0 and 4	64	2^{64}	5	R = 6

The **integral distinguishers** are the last two rows: 6 rounds with a 2^{32}-text integral, 5 rounds with a 2^{64}-text integral. These cover 4.7% and 3.9% of the 128 rounds at r0=0, leaving 122 and 123 rounds without an integral distinguisher in the tested pattern.

The first row (1 active bit, 2-text integral) is **not** a meaningful integral distinguisher: with 2 texts, "balanced output bit" is equivalent to "output bit independent of the input bit". It is reported as a **single-bit dependency propagation** result — it measures how many rounds are needed for the active bit to reach all 256 output bits (10) — but no integral security conclusion is drawn from it.

Empirical verification at r0=0. The 6-round distinguisher (word 0 active) and the 10-round single-bit dependency were verified against the real cipher (fixed random key, all non-active input bits random) over 20,000 samples each. The balanced bits predicted by the model never flipped; control bits (predicted not balanced) flipped with probabilities 6.5%–87%. The 5-round distinguisher (words 0,4 active) was not verified empirically (its integral has 2^{64} texts; a random- sample approximation with 20,000 samples is not meaningful).

Offset sweep. All 31 rotation offsets were tested at the saturation rounds of r0=0 (word0: R=6; word0,word4: R=5). The number of balanced output bits varies 8–9× across offsets (3–28 at R=5 for word0,word4; 3–26 at R=6 for word0). The offsets r0=12 and r0=27, which maximize the weight-0 linear chains, are **not** special for DP (values near the mean). The peak block is r0 ∈ [12,17], with a maximum at r0=15 (shift 16).

At r0=15 the distinguishers extend by four to five rounds over r0=0. The DP trace decays by roughly a factor of two per round:

R	6	7	8	9	10	11
word0, balanced bits	26	**12**	6	3	**1**	**0**
word0,word4, balanced bits	**15**	9	4	2	**1**	**0**

Both patterns saturate at R=11. The distinguisher covers **10 rounds** for both patterns. The tail at R=10 is a single balanced bit; the practical strength is in the 7–8 round range.

Verification at r0=15. The 12 word0 bits at R=7 and 13 of the 15 word0,word4 bits at R=6 were verified against the real cipher (20,000 samples, fixed key) as consistent with bit-0 toggle (and, for the two remaining bits word5 bit6 and word7 bit9, with simultaneous toggle of word0[0] and word4[0]). This is a necessary condition for the DP prediction, not a verification of the full $2^{32}/2^{64}$ integral. The bits at R=8–10 were not individually verified; they remain model predictions.

Comparison with Simon (with care). For Simon32/64, Todo and Morii (FSE 2016) prove a 14-round integral distinguisher with the conventional bit-based division property (31 active input bits), and a 15-round one using three subsets. 14 rounds is 44% of Simon32's 32 rounds (47% for 15 rounds). In **relative terms** MYSX resists integral distinguishers better (7.8% at the maximum round count at r0=15, 6.25% at R=8, vs 44% for Simon32), which is consistent with MYSX's more aggressive diffusion (4 parallel Feistel cells + transposition). In **absolute terms** Simon's distinguisher reaches further (14 vs 10 rounds) even though MYSX has 4× more rounds. The two are not directly comparable without normalizing by total rounds, and the MYSX analysis used only the conventional bit-based rules. (The earlier text of this paragraph quoted an 11-round Simon32 distinguisher, 34%; that figure was wrong.)

What this establishes and what it does not.

Established:

- MYSX has integral distinguishers covering 6–10 rounds depending on the rotation offset: 6 rounds (2^{32}) / 5 rounds (2^{64}) at r0=0; 10 rounds (2^{32} and 2^{64}) at r0=15, with a thin tail (1 balanced bit at R=10, 0 at R=11).
- A single active input bit stops influencing all 256 output bits after 10 rounds at r0=0.
- The standard division-property rules with the `(1,1)`-invalid XOR constraint produce the same results as the OR-conservative rule for the tested inputs.

Not established:

- Full $2^{32}/2^{64}$ integral verification at r0=15. The toggle test is a necessary condition only.
- Behaviour of the R=8–10 tail at r0=15 under a stronger verification method.
- Whether the peak at r0=15 (shift 16) reflects a structural weakness of shift-16 rounds or a property of the specific window. **This is a hypothesis, not established.**
- Behaviour for offsets outside the peak block [12,17] at extended round counts.
- Full-block inputs (255 active bits).

- The three-subset division property (introduced by Todo–Morii, FSE 2016; variant without unknown subset: Hao et al., EUROCRYPT 2020) and traceability-enhanced variants (Wang et al., 2023; citation to be verified), which may extend the distinguishers.
- Any key-recovery attack based on these distinguishers.

Historical note. The XOR rule of Todo–Morii as commonly written in the literature (`k_c = k_a XOR k_b`) is not sound when the two XOR operands share variables. In MYSX's `F = C ⊕ D`, both `C` and `D` depend on `R`, so that rule overestimates balanced bits via spurious cancellations. The first implementation of this analysis used that rule and produced oscillating, non-converging numbers (127 → 117 → 152 → ...). Those numbers are an artifact and are not reported here. The results above use the standard `(1,1)`-invalid rule, which is sound.

4.10. Beam search over the exact linear model, 128 rounds, both directions, all offsets

4.10.1. Method

Tool: `misc/mysx_linear_beam.py` (self-contained: it re-implements the exact LAT of F, the exact mask propagation of §4.3 in both directions and, for its `selftest`, the real cipher). It searches for achievable single-trail weights with `w = Σ_r Σ_p h(m_r[p])`.

Two directions.

- **Forward** (`--direction forward`, default): the offset `o` is the index of the *first* round mod 31. Starts are masks on `s[o]`, the trail covers rounds `o .. o+R-1`, and the trail must fit inside the cipher (`o + R ≤ 128`).
- **Backward** (`--direction backward`): the offset `o` is the index of the *last* round mod 31. Starts are masks on `s[e+1]` with `e = o + 31·⌊(127−o)/31⌋`, the trail covers rounds `e-R+1 .. e`, and the trail must fit inside the cipher (`e + 1 ≥ R`, i.e. `realizable_R = e + 1 ≥ R`). The propagation uses the inverse round rule of §4.3.

Merge. `python3 mysx_linear_beam.py merge a.json b.json c.json` combines any number of result files (from either direction) and reports the minimum weight per R across all offsets and both directions. The combined table of §4.10.2 is the output of `merge beam_full.json beam_pairs47.json beam_bwd.json`.

Parameters of the reported runs:

Parameter	Value
Rotation offsets	all 31 (0..30)
Rounds	up to 128
Start masks	the 256 single-bit masks of the relevant state
Beam width	2000 states per layer, ranked by (weight, Hamming weight), deduplicated by mask
Per-cell limit	`h(b) ≤ 16`; `b = 0xFFFFFFFF` (full cycle, special case of §4.2) is excluded

Parameter	Value
Valid `g` per cell	the 3 sparsest elements of the valid set (`--g-keep 3`)
Reported value	suffix minimum `ub(R) = min_{R' ≥ R} best(R')`, valid because the best weight is non-decreasing in R
Forward run	`beam_full.json`, `beam_full.log`: 8 processes, 1972 s
Backward run	`beam_bwd.json`, `beam_bwd.log`: 8 processes, 1512 s
Multi-bit run	`beam_pairs47.json`: `--starts pairs47`, did not improve on single-bit

Every exclusion or truncation above only *restricts* the search, so all reported weights remain achievable single-trail weights (upper bounds on the best weight). A larger beam, `--g-keep 0` and multi-bit starts can only lower them.

Validation of the tool, before the runs:

- `selftest` passes in both directions: (1) the closed-form LAT of F against sampling, (2) one round of the model against the real round function with random keys, (3) the 9-round weight-0 chains at r0 = 12 and r0 = 27 found by the forward search and confirmed with `|corr| = 1` on the real cipher (256 random states per chain), (4) every forward edge is a backward edge with the same cost (325 edges checked), (5) the 9-round weight-0 chains are found backward at end residues 20 and 4 with `|corr| = 1` on the real cipher, (6) a backward R=14 trail with weight 21 is a valid forward trail with equal weight.
- `reproduce` returns exactly the §4.5 table for r0 = 12 and r0 = 27 (R ≤ 14, beam 1200, `h ≤ 2`).

4.10.2. Results

Best achievable single-trail weight per number of rounds, minimum over offsets and both directions, subject to the trail fitting inside the 128-round cipher (forward: `r0 + R ≤ 128`; backward: `realizable_R = e + 1 ≥ R`):

R	best weight	offset	direction	≈ texts 2^{2w}
9	0	4	backward	2^{0}
10	1	4	backward	2^{2}
12	2	20	backward	2^{4}
14	5	20	backward	2^{10}
16	8 †	23	forward	2^{16} †
18	24	20	backward	2^{48}
20	38	20	backward	2^{76}
22	63	23	forward	2^{126}
24	88	3	backward	2^{176}

R	best weight	offset	direction	≈ texts 2^{2w}
32	204	3	backward	2^{408}
40	361	20	backward	2^{722}
48	485	20	backward	2^{970}
56	630	27	forward	2^{1260}
64	786	27	forward	2^{1572}
80	1081	9	forward	2^{2162}
96	1336	21	backward	2^{2672}
112	1671	21	backward	2^{3342}
128	2248	3	backward	2^{4496}

† The weight-8 trail of this row cancels once its hull is computed (four trails of weight 8 with the same end masks sum to zero, §4.10.5), so 2^{16} texts is not a distinguisher cost. The best certified correlation at R=16 is the offset-20 backward hull, $|\text{corr}| = 2^{-11}$ ($\approx 2^{22}$ texts). The same caveat applies to every row: a row is an achievable *single-trail* weight, not a correlation.

Largest R with an achievable single trail of weight < 32 ($\approx 2^{64}$ texts): **19**. Largest R with weight < 64 ($\approx 2^{128}$ texts): **22** (weight 63 at R=22, offset 23 forward; 49 at R=21). This is a **tie with the threshold**: at beam 1000 the same cell gives 64/71, so the defensible statement is "≈20–22 rounds", not "22".

Direction alternates with R. Among the tabulated round counts the minimum is reached by a backward trail at R=9, 10, 12, 14 (tied with offset 23 forward), 18, 20, 24, 32, 40, 48, 96, 112 and 128, and by a forward trail at R=16, 22, 56, 64 and 80. The two directions align with different sets of rotation windows, so neither is uniformly better. The combined table is the right object; do not compare the forward-only and backward-only tables row by row.

R=128 is a real trail. In forward mode only r0=0 fits a 128-round trail (`o + R = 128`). In backward mode, offset 3 has `e = 3 + 31·⌊124/31⌋ = 127`, `realizable_R = 128`, and the trail covers rounds 0..127. The combined R=128 = 2248 at offset 3 backward is a **real trail inside the cipher**, not a virtual extension. Offsets with `realizable_R < R` are excluded from the minimum (the tool checks this explicitly). The `v0.1-draft` statement that "for r0>0 the value at R=128 is a virtual extension" was forward-specific; it does not hold for backward offset 3.

Per-offset detail at R=14 (single-bit starts, beam 2000, merged). For compactness, the backward values are shown (they include the best cell of the whole table, offset 20 = 5); the forward values are in `beam_full.json` and the merged per-offset arrays are in the output of `merge`.

offset (backward, last round mod 31)	0	1	2	3	4	5	6	7	8	9	10
weight	17	11	11	8	11	36	61	46	34	47	43

offset	11	12	13	14	15	16	17	18	19	20	21
weight	24	53	42	38	31	24	16	12	8	**5**	21

offset	22	23	24	25	26	27	28	29	30
weight	25	21	13	19	41	36	36	28	26

The best cell at R=14 is offset 20 backward (weight 5). The next best are offsets 3 and 19 backward (weight 8). The forward per-offset table is available in `beam_full.json`; the best forward cell at R=14 is offset 23 (weight 5), tied with offset 20 backward. The spread (5 to 61 across offsets for the same R) is the evidence that the rotation offset matters a great deal, which is also why the differential analysis of §3 needs to be repeated over offsets.

Growth. Between R=22 and R=112 the best weight grows by roughly 12–21 per round (12.5 for 22→24, 14.5 for 24→32, 19.6 for 32→40, 15.5 for 40→48, 18.1 for 48→56, 19.5 for 56→64, 18.4 for 64→80, 15.9 for 80→96, 20.9 for 96→112). The last step, 112→128, is ≈36 per round (1671 → 2248). That step is inflated by the fitting rule: only offset 3 backward fits a full 128-round trail, so the R=128 value is not a minimum over all 31 offsets as the others are. The average between R=48 and R=128 is ≈22 per round. The weight never settles at a low per-round cost. Note also that single-trail weights above ≈256 say nothing about the cipher: with a 256-bit block the sum over the many trails of a hull is what matters there, and no hull-aware statement is made beyond R≈14.

4.10.3. Hull check of the trails found

A single-trail weight `w` predicts `|corr| = 2^{-w}` only when the trail is alone in its hull. The beam trails were therefore checked against the real cipher (fixed key, final whitening disabled, 2^{17}–2^{18} random states), prefix by prefix, using the stored per-round masks. Script: `misc/verify_beam_trail.py`; trail counting: `misc/hull_count.py`.

Trail of offset 20 backward, R=14, w=5 (covers rounds 100..113; `first_round[14] = 100`). Key seed 1, 2^{17} states ($\sigma = 0.0028$):

R	trail weight w	predicted 2^{−w}	empirical corr
1	1	0.5000	−0.4947
2	3	0.1250	−0.1206
3	4	0.0625	−0.0601
4	4	0.0625	+0.0601
5	5	0.0312	+0.0279
6	5	0.0312	−0.0279
7	5	0.0312	−0.0279
8	5	0.0312	−0.0279
9	5	0.0312	−0.0279
10	5	0.0312	+0.0279
11	5	0.0312	+0.0279
12	5	0.0312	−0.0279
13	5	0.0312	+0.0279
14	5	0.0312	−0.0279

`hull_count` on the saved end mask returns **exactly one trail** at the minimum weight for every R from 1 to 14 (layer sizes 4, 64, 84, 10, 4, 2, 1, 1, …, 1; the layer collapses to a single trail from R=7). The predicted and empirical magnitudes agree within σ at every R. There is **no cancellation** for this end mask. **This is a verified linear distinguisher on a 14-round window (rounds 100–113).**

How to read it. The weight reaches 5 at R=5 and stays there up to R=14: the trail is five cheap rounds followed by a 9-round weight-0 chain (the free chain of §4.3). The rows R=5…14 are therefore the same signal observed after more free rounds, not ten independent confirmations. The trail lies at rounds 100–113, so it is a property of that window of the cipher; it is not a distinguisher for a 14-round version of the cipher starting at round 0, and it is not an attack on the 128-round cipher. Section 4.10.5 extends the same trail to R=16 by two more weighted rounds (R=15: weight 8, R=16: weight 12).

Trail of r0=23 forward, R=14, w=5 (start bit 197 of `s[23]`; end masks bits {46, 66, 84, 196} of `s[37]`), key seed 1, 2^{17} states (σ = 0.0028):

R	trail weight w	predicted 2^{−w}	empirical corr
1–4	0	1	±1.0000
5–7	1	0.5000	−0.5014
8–10	2	0.2500	±0.0010
11–13	3	0.1250	±0.0012
14	5	0.0313	−0.0001

The model is exact up to R=7. At R=8 the empirical correlation is zero instead of ±1/4. An exact enumeration of all trails with weight ≤ 4 from the same start mask shows why: **two distinct trails of weight 2 reach the same end masks at R=8, 9 and 10** (and one at R=7), so their contributions cancel. Two independent keys at R=14 (2^{18} states each, σ = 0.002) give −0.0004 and +0.00001 against a single-trail

prediction of 0.031. For this trail the single-trail weight is therefore not a usable estimate of the correlation.

Other offsets, R=14 trails, one key (seed 1), 2^{16} states (σ = 0.0039):

r0	direction	R	single-trail 2^{−w}	empirical corr	relation
12	forward	10	0.5000	+0.4969	equal
12	forward	12	0.1250	−0.1161	equal (sign key-dependent)
27	forward	10	0.1250	+0.2503	×2
27	forward	12	0.0156	+0.0369	×2.4
26	forward	10	0.1250	+0.2412	×2
26	forward	12	0.0625	+0.1218	×2
24	forward	10	0.1250	+0.0049	cancelled
24	forward	12	0.1250	−0.0049	cancelled

At R=14 the predicted correlations (2^{-7} to 2^{-12}) are below the resolution of this test (σ = 0.0039) for these four forward trails, so no statement is made at R=14 for them. The offset-20 backward trail (previous table) is measurable at R=14 because its weight stays at 5 from R=5 to R=14, so the test at R=14 is the same signal as at R=5 (where σ is adequate).

Reading. Across the trails checked, the real correlation equals the single-trail value (offset 20 backward at R=14; r0=12 forward up to R=12), is about twice as large (r0=26, r0=27 forward; a second trail adds constructively), or is exactly cancelled (r0=23 forward from R=8; r0=24 forward). The observed gain is at most ≈2.4× (about 1.3 bits); the observed loss can be total. Hull effects are therefore not a uniform correction, and they are not always destructive: some trails survive intact and are genuine distinguishers of their window.

4.10.4. What §4.10 establishes and what it does not

Established:

- In the single-trail model, and within the searched family (single-bit starts, beam 2000, 3 sparsest g per cell, both directions), no trail with weight < 64 (≈2^{128} texts) was found for more than 22 rounds, and none with weight < 32 for more than 19 rounds. The threshold at 22 is a tie (weight 63): "≈20–22" is the defensible reading.
- The best weight grows by roughly 12–21 per round from R≈22 to R=112 (≈22 on average from R=48 to R=128, inflated by the last step).
- The cheapest trails alternate direction (see §4.10.2 for the exact R values). The best cell at R=14 is offset-20 backward (weight 5), tied with offset-23 forward.
- The tool agrees with the real cipher wherever a trail is alone in its hull (offset-20 backward up to R=14; r0=12 forward up to R=12; r0=23 forward up to R=7).
- Single-trail weight is not a reliable proxy for correlation in general: exact cancellation was observed (r0=23 and r0=24 forward).
- The offset-20 backward trail at R=14 is isolated in its hull (`hull_count` = 1 trail) and its empirical correlation matches 2^{-5} on the real cipher. The exact budgeted

hull of §4.10.5 extends it to R=16 (rounds 98–113, $|\text{corr}| = 2^{-11}$, two trails of weight 12 adding, empirical $6.1\times10^{-4} \pm 6.1\times10^{-5}$ against a hull prediction of 4.9×10^{-4}). **This is a verified linear distinguisher on a 16-round window (rounds 98–113).**

- At R=17 the best end pairs of offset 20 backward (7 of 8 computed) and offset 23 forward (8 of 8) cancel exactly within the enumerated budget, for both keys tested (§4.10.5).
- Multi-bit starts (pairs47, rand:5000:4) did not improve on single-bit starts at any R. pairs47 is a small structured subset; rand:5000:4 covers 5000 masks out of 2^{256} and is not conclusive for structured multi-bit masks.

Not established:

- A **distinguisher length beyond R=16**. R=16 is established for the offset-20 backward trail (exact hull over trails of weight ≤ 14 plus a one-key measurement). At R=17 the best pairs cancel within the budget, which does not exclude a smaller non-zero correlation from heavier trails or a distinguisher from other end pairs. R≥18 is undecided: the weights are ≥ 24 and the frontiers of the enumeration exceed the state caps (§4.10.5). The trails of §4.10.2 at R=18–22, including the weight-63 trail at R=22 that sits at the 2^{128} threshold, have not been checked.
- A **lower bound**. The beam gives achievable weights only. The truncations (`g-keep 3`, beam, single-bit starts, exclusion of `b = 0xFFFFFFFF`) may hide cheaper trails.
- That hull gain stays small at larger R. At R=24 (w=88) a gain of $\approx 2^{24}$ in correlation ($88 - 64$) would be needed to bring the data cost down to the 2^{128} budget; the gains observed so far are ≤ 2.4× (exactly 2× at R=15 and R=16) and were measured only up to R=16.
- That multi-bit starts do not help at all. The tested samples (pairs47, rand:5000:4) do not improve on single-bit, but the random sample is too small to rule out structured multi-bit masks that might. The honest statement is “the multi-bit starts tested do not improve on single-bit”.
- Key recovery from any hull, differential analogues, and the effect of the final whitening and of key-recovery rounds around a distinguisher.

4.10.5. Exact hull at R=15–17 (budgeted, meet-in-the-middle)

§4.10.4 named the exact signed hull of the best end-mask pairs at R≥16 as the next step. This section does it for R=15–17 and reports where it stops. Tools: `misc/hull_bidir.py`, `misc/scan_endpairs.py`, `misc/test_signed_dp.py`, `misc/emp_check.py`. Inputs: `misc/beam_paths_bwd20.json` and `misc/beam_paths_fwd23.json`, re-runs of `mysx_linear_beam.py run` for offset 20 backward and offset 23 forward with `--paths 15..22` (their `ub` values and the R=14/20/24 paths are identical to `beam_bwd.json` / `beam_full.json`). A summary with the reproduction commands is in `misc/HULL_R15_R17.md`.

Method. `hull_count.py` expands only forward from the start mask, so its layers explode when heavy rounds sit at the start of the trail (backward trails). `hull_bidir.py` grows a forward frontier from the start mask `u` and a backward frontier from the

end mask v (always extending the smaller one), prunes each state with the minimum weight already paid by the opposite frontier, and streams the last round into the join. Any trail u → v of total weight ≤ wmax has prefix and suffix weight ≤ wmax, so the result is **complete for weight ≤ wmax** (cells with b = 0xFFFFFFFF are skipped, which matters only from weight 15 per cell). With --keys K it also sums the exact signed contributions (cell sign from the LAT, key sign (−1)^{m·rk}) for K random keys.

Validation. mysx_linear_beam.py selftest passes; the signed hull equals propagate() of mysx_linear_model.py on 40 of 40 end masks (windows r0=12 with R=10, 11, 12 and r0=27 with R=10, two keys; test_signed_dp.py); at R=14, offset 20 backward, it reproduces hull_count.py (one trail of weight 5). These validation windows are forward and R ≤ 12; the signs of the backward step are validated indirectly by the empirical measurements below.

Results. w1 is the single-trail weight of the beam trail; "budget" is wmax (all trails up to it are enumerated). The hull value is the same for keys rand_key(1000) and rand_key(1001).

trail	R	rounds	w1	trails u→v found	budget	exact hull
offset 20 bwd	14	100–113	5	1 × w5	7	$\pm 2^{-5}$ (isolated)
offset 20 bwd	15	99–113	8	2 × w8	10	$+2^{-7}$ (two trails add)
offset 20 bwd	16	98–113	12	2 × w12	14	$+2^{-11}$ (two trails add)
offset 20 bwd	17	97–113	16	2 × w16	18	0 (exact cancellation)
offset 23 fwd	15	23–37	8	4 × w8	10	0
offset 23 fwd	16	23–38	8	4 × w8	10	0
offset 23 fwd	17	23–39	14	4 × w14	16	0

Empirical check against the real cipher (fixed key rand_key(1000)):

- offset 20 backward, R=16, N = 2^{28}: corr = $+6.06\times10^{-4} \pm 6.1\times10^{-5}$. The hull predicts 4.88×10^{-4} (+1.9σ); the single-trail value would be 2.44×10^{-4} (+5.9σ) and zero is +9.9σ away. The sign matches the model.
- offset 23 forward, R=16, N = 2^{24}: corr = $+3.7\times10^{-4} \pm 2.4\times10^{-4}$ (1.5σ from zero); the single-trail value would be $\pm 3.9\times10^{-3}$ (16σ away). Consistent with the cancellation.

R=17 scan (scan_endpairs.py --R 17, scan_R17.log): the top 8 end pairs of offset 20 backward (weight 16; 7 computed, 1 exceeded the state cap of 800,000) and of

offset 23 forward (weight 14; 8 of 8) all give an exact hull of 0, for both keys, within a budget equal to the pair's single-trail weight. Offsets 19 backward, 24 forward, 3 backward and 2 backward have R=17 weights 23–26 and exceeded the state cap: undecided.

Reading.

- The verified window of the offset-20 backward trail grows from 14 to 16 rounds (rounds 98–113). The weights at R=14, 15, 16, 17 are 5, 8, 12, 16: the two added rounds cost 3 and 4, so the trail is seven weighted rounds (98–104, total weight 12) followed by the 9-round weight-0 chain (105–113). At R=15 and R=16 the hull is a factor 2 above the single-trail value because two trails of equal weight add.
- The R=16 row of the §4.10.2 table (weight 8, offset 23 forward) is the trail that cancels; its "2^{16} texts" is not a distinguisher cost.
- At R=17 the best pairs of both offsets cancel exactly within the budget, for both keys, in all 14 (pair, key) cases of offset 20 backward and all 16 of offset 23 forward, and at R=15–17 the two keys agree in every case. This suggests that the relative sign of the paired trails does not depend on the key, i.e. that the addition or cancellation is structural; two keys are too few to assert it.

What this does not establish.

- A hull of 0 means "no net contribution from trails up to the budget", not a correlation of exactly 0. Trails above the budget are not summed (weight ≥ 15 at R=16, ≥ 19 at R=17 in the main runs; the R=17 scan used a budget equal to the pair's own weight). The measured correlation at R=16 is 1.9σ above the hull prediction, which is compatible with noise and also with a few trails of weight 15 (each adds 2^{-15}); a run at R=16 with `--wmax 18` and more keys has not been done.
- R≥18 is undecided: the weights are ≥ 24 and the frontiers exceed 2.5 million states. Only end pairs returned by the beam were tried; other pairs at R=17–22 may not cancel. Covering them needs a different method (the SAT/MILP hull encoding of `LINEAR_MODEL.md` §7) or a much larger scan. In particular the weight-63 trail of the R=22 row of the §4.10.2 table, which decides the position of the 2^{128} threshold, is untouched.
- The sign was checked empirically for one key only.
- None of this is a key-recovery cost, a lower bound, or an attack on the 128-round cipher.

4.11. Statistical randomness of the CTR keystream

Tool: NIST Statistical Test Suite (STS) 2.1.2, plus the generator script `misc/gen_nist_input.js`. This section reports the first statistical randomness test of the MYSX keystream; it was absent from `v0.1-draft`.

Method. One hundred independent sequences of 1,000,000 bits each were produced as follows: for each sequence, a fresh uniformly random 512-bit key and a fresh uniformly random 128-bit nonce were drawn, an all-zero plaintext of 125,000 bytes was encrypted with the CTR mode of §2.8 (`withFinalWhiten = true`, i.e. the full 128-

round cipher), and the resulting keystream was written as a raw binary file. The 100 files (12,500,000 bytes in total, exactly 100,000,000 bits) were concatenated and fed to STS 2.1.2 with `./assess 1000000`, all 15 tests enabled, default parameters (block frequency M=128, template m=9, approximate entropy m=10, serial m=16, linear complexity M=500), binary input mode, and 100 bitstreams.

Results. No test failed. No row is flagged with * or `FAIL`, no uniformity p-value is below the 0.0001 threshold, and no proportion is below the NIST minimum for its sample size.

Test	Proportion	Uniformity p-value	Verdict
Frequency	97/100	0.851383	pass
Block Frequency	97/100	0.289667	pass
Cumulative Sums (fwd/rev)	97/100	0.350485 / 0.637119	pass
Runs	100/100	0.834308	pass
Longest Run of Ones	100/100	0.096578	pass
Rank	100/100	0.574903	pass
Discrete Fourier Transform	96/100	0.779188	pass
Non-Overlapping Template (148 sub-tests)	96–100/100	min 0.017912	pass
Overlapping Template	99/100	0.978072	pass
Universal	99/100	0.051942	pass
Approximate Entropy	100/100	0.719747	pass
Random Excursions (8 states, n=65)	64–65/65	min 0.041438	pass
Random Excursions Variant (18 states, n=65)	63–65/65	min 0.063482	pass
Serial (two sub-tests)	100/100	0.798139 / 0.366918	pass
Linear Complexity	100/100	0.224821	pass

The NIST minimum pass rate for a sample of 100 sequences is approximately 96/100 (61/65 for the random-excursion tests, which use 65 sequences because of cycle counts). The tightest margins in this run are the Discrete Fourier Transform test and two Non-Overlapping Template sub-tests, all at exactly 96/100; every other 100-

sequence test is at 97/100 or above. The smallest uniformity p-value in the whole report is 0.017912, roughly 180× the 0.0001 rejection threshold.

What this establishes and what it does not.

Established:

- The CTR keystream of the full 128-round MYSX passes the complete NIST SP 800-22 battery on 100 independent sequences of 10^6 bits, with fresh keys and nonces per sequence, no failures, and margins inside the NIST acceptance region for every test.

Not established:

- Any statement about cryptographic security. SP 800-22 detects gross statistical biases (non-uniformity, short-period artefacts, linear structure at the bit level, excess runs); it does not detect differential, linear-hull, or algebraic weaknesses, and a primitive can pass it and still be trivially broken.
- Any statement about reduced-round versions of the cipher. The test covers the full 128 rounds with the final whitening enabled, exactly as the public API uses it.
- Any statement about the AEAD construction as a whole (HMAC-SHA256 tag, packet format, nonce policy); only the raw keystream was tested.
- Independence between the 100 sequences beyond the fact that each uses a fresh random key and nonce. The sequences were drawn independently; no pairwise statistical test was run.
- Resistance to `TestU01` (`Crush`/`BigCrush`) or `dieharder`, which were not run. `TestU01 BigCrush` is strictly stronger than SP 800-22 and would be the natural next step; it is left for future work.

The purpose of this section is narrow and negative: it rules out the trivial failure mode of a keystream with detectable statistical structure. It is not evidence of security, and it does not change the verdict of §10.

5. Related-key cryptanalysis

Tools: `misc/keyschedule_test.js` (basic) + `misc/keyschedule_test_extended.js` (22 tests). Coverage: exhaustive weight-1 (512 bits), exhaustive weight-2 (130,816 pairs), weight-3 “same Δ” (1,680 trials), beam search up to weight-6 (beam 50), 5 base keys, structured patterns, cycles, weak keys.

Results (summary):

- No persistent `Δrk[r]=0` for r≥1 in any test.
- Beam search score: 139 (ideal 128).
- No weak keys, cycles, or fixed points.
- Slow diffusion at r=1–4 (expected: one Feistel round per cipher iteration).
- Worst bit (minimum weight at r=1): bit 160 (word 5, bit 0), 1-bit difference. Consistent across 5 base keys (structural, not noise).

Limitations. These are heuristic tests, not a proof. We did not run a targeted

MILP/SAT related-key model (~650k variables); adaptive related keys, biclique attacks, and algebraic attacks on the key schedule are not covered. **MYSX's related-key security is unknown.**

Round-key recovery. The only non-linear operation of the key schedule is the 8 modular additions per step; everything else is XOR and rotation. The schedule is bijective, so any 16 consecutive state words determine the master key. How many round keys must be known for a SAT solver to recover the master key (an estimate by counting: about three consecutive `rk[r]`) has not been measured. This number bounds what a partial key-recovery attack on a few rounds can lead to.

6. Generic attacks

Attack	Complexity	Verdict
Classical brute force	2^{512} operations (uniformly random 512-bit key)	Infeasible
Grover (quantum)	2^{256} iterations (uniformly random 512-bit key)	Infeasible with foreseeable technology
Block collisions (birthday)	2^{128} blocks	Infeasible

Note on Grover. With a uniformly random 512-bit key, Grover leaves 256 bits of effective security. With 1,000,000 logical qubits at 1 ns/iteration, the time would be $\sim 10^{60}$ years. This is a theoretical estimate; no post-quantum analysis of the cipher beyond Grover has been done.

Note on the 512-bit key. The key-derivation functions of Section 2.10 (and therefore every key produced by `deriveAeadKeys` and `deriveAeadKeysFromPassword`) start from HKDF-Extract, whose output is a 256-bit pseudorandom key (HMAC-SHA256). HKDF-Expand can output 64 bytes, but they are a deterministic function of that 256-bit value, so the entropy of a derived 512-bit `encKey` is at most 256 bits. If the master secret has 256 bits of entropy, the effective classical security is 256 bits (not 512) and the Grover figure is 2^{128} iterations (not 2^{256}). The "key size" of MYSX is not the security level of the system. The 2^{512} and 2^{256} figures apply only if the caller supplies a uniformly random 64-byte key directly to `aeadEncrypt`.

7. Simon cryptanalysis: what transfers and what does not

MYSX uses Simon's `F`. It is **not** Simon. The differences:

- Block size and branch organization: MYSX is an SPN-like structure with four parallel Feistel cells over 32-bit word pairs (Section 2.1). Simon is a classical 2-branch Feistel over `n`-bit halves.
- Four parallel `F` per round (Simon uses 1).
- Myszkowski byte transposition (Simon has none).
- Round-dependent word rotations (Simon has none).
- Non-linear ARX key schedule (Simon's is linear).

- 128 rounds vs 68.

What transfers. The exact LAT of F — its closed form $|corr| = 2^{-h(b)}$, its validity condition $l \in colspace(A_b)$, and its sign structure — is a property of the function and therefore applies to MYSX regardless of the surrounding structure (Section 4.2 is the transfer). That is the correct starting point for MYSX-specific linear cryptanalysis; the failed models of the v0.0 draft used a different (incorrect) F model.

What does not transfer. Key recovery from Simon's linear key schedule does not transfer: MYSX's key schedule is non-linear. Linear hull attacks against Simon are highly structured around Simon's Feistel-branch and its round-reduced behaviour; the MYSX hull is a different object and must be computed from the MYSX round model (Section 4.6). Integral distinguishers against Simon rely on Simon's particular 2-branch Feistel structure and its key schedule; the MYSX integral distinguishers in §4.9 are considerably shorter, and their relationship to Simon's is one of normalized coverage, not direct transfer.

Relevant literature on Simon (cited for context; the figures below were checked against the papers' abstracts, other figures quoted in earlier drafts were not verified and have been removed):

- Linear-hull key-recovery attacks with dynamic key-guessing reaching 49 rounds of Simon128/128 (Chen–Wang, "Improved Linear Hull Attack on Round-Reduced Simon with Dynamic Key-guessing Techniques", FSE 2016, ePrint 2015/666).
- Improved linear characteristics and hulls, including a 34-round linear characteristic with correlation 2^{-61} for Simon128 (Shi–Hu–Sun–Song– Qiao–Ma, "Improved Linear (hull) Cryptanalysis of Round-reduced Versions of SIMON", ePrint 2014/973).
- Differential analysis of Simon and Speck (Biryukov–Roy–Velichkov, "Differential Analysis of Block Ciphers SIMON and SPECK", FSE 2014).
- Integral distinguishers: 14 rounds for Simon32 with the conventional bit-based division property, and 15 rounds with three subsets (Todo–Morii, "Bit-Based Division Property and Application to Simon Family", FSE 2016); 14-, 16-, 18-, 22- and 26-round distinguishers for Simon32/48/64/96/128 (Xiang–Zhang–Bao–Lin, ASIACRYPT 2016).
- Structural observations on the SIMON family (Kölbl–Leander–Tiessen, "Observations on the SIMON block cipher family", CRYPTO 2015).

None of these results is transferred to MYSX in this paper. The MYSX-specific analysis (Sections 3, 4, 5) is what provides the evidence; Simon attacks are context, not direct transfer.

8. Analysis limitations

- **Differential**: single-trail upper bounds over 31 rotation offsets (inputs chosen by the free chain table at r0=10, or w2b30 for r0=0); two independent deterministic chains verified empirically (13 rounds at r0=10, 6 rounds at r0=15); no lower bounds; no full clustering analysis at r0≠0; the exact maximum of weight-0 chains (13 rounds) is proven, but the non-zero-weight tables are upper bounds

from a beam or time-limited MILP.

- **Linear**: an exact model of `F` is established and matches the real cipher on small round counts, including exact signed hulls at r=10, 11, 12. A beam search over the exact model, **in both directions and merged over all 31 offsets**, up to 128 rounds (§4.10), gives achievable single-trail weights only: 2 at R=12, 5 at R=14, 38 at R=20, weight < 64 up to R=22 (63, a tie) and 2248 at R=128 (offset 3 backward; a real trail inside the cipher). The offset-20 backward trail at R=14 (rounds 100–113) is **isolated in its hull** (`hull_count` = 1) and its empirical correlation matches 2^{-5}; an exact budgeted hull extends it to R=16 (rounds 98–113, |corr| = 2^{-11}): that is a verified linear distinguisher on a 16-round window. Other trails cancel or are ≈2× larger (at R=17 the best pairs cancel within the budget), so single-trail weights are not universally distinguisher lengths. No lower bound over 128 rounds; no hull-aware bound beyond R=17; hulls are exact only up to a weight budget; `g` choices truncated to the 3 sparsest per cell; no key-recovery cost estimate. Multi-bit starts (pairs47, rand:5000:4) were explored and did not improve on single-bit starts, but the random sample is too small to be conclusive for structured multi-bit masks.
- **Integral**: 6–10 round distinguishers depending on the rotation offset: 6 rounds (2^{32}) / 5 rounds (2^{64}) at r0=0; 10 rounds (2^{32} and 2^{64}) at r0=15, with a thin tail (1 balanced bit at R=10, 0 at R=11). The offset sweep shows the round count is **offset-dependent**. Verification covers the R=7/R=6 bits at r0=15 (consistent with bit-0 or double toggle; not full 2^{32}/2^{64} integrals); the R=8–10 tail is a model prediction. Only r0=0 was tested for three-subset and traceability-enhanced DP; full-block inputs and key-recovery attacks were not explored.
- **Related-key**: heuristic tests up to weight-6; no targeted MILP; number of round keys needed to recover the master key not measured.
- **Impossible differential, boomerang, algebraic**: not analyzed.
- **No independent public cryptanalysis.**
- **No post-quantum analysis** beyond the trivial Grover bound.
- **No side-channel analysis.**
- **Functional tests** (`mysx.test.js`, 17 tests) cover round-trip, AEAD negatives, nonce policy, PBKDF2, avalanche. They do not prove security.

9. Engineering notes

9.1. API surface

- Public API: `aeadEncrypt`, `aeadDecrypt`, `deriveAeadKeys`, `deriveAeadKeysFromPassword`, `NonceGuard`.
- Test API: **disabled by default**. Enabled only when the environment variable `MYSX_ENABLE_TEST=1` is set in the process before requiring the module. There is no code-level toggle.
- CTR (`_ctrXor`) is not exported from the public API.

9.2. NonceGuard

`NonceGuard` is a best-effort defense against nonce reuse. It:

- counts messages per key (default 2^{32});
- detects recent nonce reuse (window of 1024 per key);
- caps the number of tracked keys (default 1024, LRU eviction);
- identifies keys by a non-cryptographic 64-bit identifier made of two independent 32-bit lanes (a 32-bit FNV-1a lane and a multiplicative lane), **not** by the raw key bytes; `onSave(state)` therefore does not leak the secret key;
- throttles `onSave` calls (default: once every 100 `check()` calls) to avoid per-message state copies;
- **does not persist across restarts by itself**;
- **does not share state across processes or devices**.

Persistence is provided by `onSave(state)` and `initialState` at construction, or by `save()` / `load(state)` for manual control.

It is not a substitute for key lifecycle management. If the application reuses `(key, nonce)` across restarts, CTR confidentiality is broken. When a key is evicted from the tracked set its counter and nonce window are lost. See `mysx.prod.js` for the exact contract.

What it actually buys. `aeadEncrypt` draws a fresh 16-byte random nonce for every message, so an accidental repeat of `(key, nonce)` has probability about 2^{-65} even after 2^{32} messages under one key (birthday bound on 128 bits). The nonce-window check therefore guards mainly against a faulty random-number generator; the part that matters in practice is the per-key message limit, which is also lost when a key is evicted or the process restarts. The option name `unsafeNoGuard` reflects that the guard is missing, not that nonces are imposed by the caller (they are not: there is no public path to impose a nonce).

9.3. aeadEncrypt size validation

`aeadEncrypt` validates plaintext size **before** allocating the CTR output buffer, and enforces a configurable `MAX_PLAINTEXT_BYTES` limit (aligned with `MAX_CT_BYTES`, $2^{31}-1$). Applications that require larger limits should raise both limits explicitly and manage memory accordingly.

9.4. Type validation

All public functions validate that byte inputs are `Uint8Array` of the expected length (where applicable). Passing a `string` as `aad` or `plaintext` throws instead of silently coercing to array-like behavior.

9.5. KAT digests

Two digests are documented in `README.md`:

- `KAT.json` public digest: SHA-256 over the full JSON (block + AEAD + KDF). This is the artifact a third-party implementation should reproduce.
- `KNOWN_KAT_DIGEST` (internal): SHA-256 over 16 blocks with `key[i] = i`. Used as fast regression in `mysx.test.js`. It is strictly a subset of the public KAT, not a different KAT.

A third-party implementation should target the **public digest**.

9.6. Harness validation

The linear sampling harness is validated against a hand-computed case at w=8 (r=1, α = {word0 bit2, word4 bit0}, β = word0 bit3, fixed key). The linear approximation $x_0 \oplus F(x)_2 = x_1 x_6$ has correlation exactly ±1/2 with fixed key. Observed correlation: **0.4993** (expected 0.5, within sampling noise). With random keys per sample, the sign of the correlation flips per sample and the observed |corr| drops to **0.0022** (expected ≈ 0).

This confirms the harness is correct when used with fixed keys. It also confirms that the practice of "random keys per sample", used in an earlier version of `misc/mini_trail.js`, washes out genuine linear signals and must not be used for linear cryptanalysis.

See `misc/validate_lineal_harness.js`.

The mini-MYSX with `w=8, kw='MYSZKOWS'` is **degenerate**: the permutation becomes `[4,0,5,2,7,6,1,3]`, the L→R→L cycle at word 0 means a single-bit difference never passes through F, and the differential probability is 1 at all rounds. This is not representative of MYSX. Use `w=8, kw='MYSZ'` (non-degenerate from r=3) or w=32 directly. See `misc/check_mini_nondegenerate.js`.

The integral division-property harness (`misc/verify_division.py`) is validated against the R=10 single-bit dependency: the two bits the model predicts as balanced (word 0 bit 31, word 3 bit 6) are confirmed balanced over 20,000 random samples, and the other 62 bits of words 0 and 3 are confirmed not balanced. See `misc/verify_division.py` and §4.9.

The beam-search tool (`misc/mysx_linear_beam.py`) has its own `selftest` (LAT of F, one round against the real round function, 9-round weight-0 chains against the real cipher, forward/backward edge duality, backward 9-round weight-0 chains, backward R=14 trail is a valid forward trail) and a `reproduce` command that regenerates the §4.5 table. See §4.10.1.

The differential free chain table (`misc/diff_free_chain_table.py`) was verified against the MILP for the offsets tested; the maximum weight-0 chain length (13 rounds) was verified by exhaustive linear algebra over 2^{128} left-half differences (external review). See §3.5.

The differential verification scripts (`misc/verify_diff_trail.py`, `misc/verify_diff_13r.py`) check the deterministic trails of §3.5 against the real cipher at 100% propagation over 262,144 pairs × 8 keys.

10. Verdict

MYSX-256/512 v0.1-draft2 is an experimental proposal, not a secure cipher. The differential analysis (§3) reports upper bounds on single-trail weights over all 31 rotation offsets, not a security proof. Probability-1 differential weight-0 chains exist at

some offsets; the longest, 13 rounds at r0=10 (rounds 10–22 mod 31), is the maximum possible, verified by exact linear algebra and empirically with 100% propagation. The best single-trail weight found at R=14 ranges from 2 (r0=10) to about 120 across offsets; the 128-bit threshold is reached at R≈24 for r0=10 versus r≈15 for r0=0. The r=4 trail at r0=0 has been directly verified with clustering factor 2.00×. The linear analysis has a correct, exact model of `F` and of the round; it reproduces the real cipher's linear behaviour on the small round counts where it has been checked (verified `|corr| = 1` chains at 5 and 9 rounds; exact signed hulls at r = 10, 11, 12). A beam search over that model, run in **both directions** for all rotation offsets, up to 128 rounds, and merged, finds that the cheap region ends at roughly 20 rounds (single-trail weight 38 at R=20, below 64 only up to R=22 with weight 63 — a tie with the threshold — then growing by roughly 12–21 per round up to R=112; 2248 at R=128, offset 3 backward, which is a real trail inside the cipher). The offset-20 backward trail at R=14 (weight 5, rounds 100–113) is **isolated in its hull** (`hull_count = 1`) and its empirical correlation matches the prediction, and an exact budgeted hull extends it to R=16 (rounds 98–113, $|\text{corr}| = 2^{-11}$): **this is a verified linear distinguisher on a 16-round window**, made of seven weighted rounds and a 9-round weight-0 chain. At R=17 the best end pairs cancel within the enumerated budget. For other trails, a check against the real cipher shows that a single-trail weight can be exact, ≈2× too low, or exactly cancelled, so these numbers are evidence about where cheap trails end, not universally how long a distinguisher is. No lower bound over 128 rounds has been established, no key-recovery attack has been attempted, and hull-aware bounds beyond R=17 are unexplored. Multi-bit starts (pairs47, rand:5000:4) did not improve on single-bit starts, but the random sample is too small to be conclusive. The integral analysis, extended to all 31 rotation offsets, establishes 6-round (2^{32} texts) and 5-round (2^{64} texts) distinguishers at r0=0, extended to 10 rounds (both patterns) at r0=15, with a thin tail (one balanced bit at R=10) and practical strength in the 7–8 round range. The R=7/R=6 bits are consistent with a 2-text or 4-text toggle; the full $2^{32}/2^{64}$ integrals were not verified. The offsets r0=12 and r0=27 are not special for DP. **The three analysis directions peak at different offsets** (differential r0=10, linear free chains r0=12/27, linear hull at offset 20 backward ≈ r0=5–6, integral r0=15), so none of the single-offset bounds previously reported describes the cipher as a whole. The CTR keystream passes the full NIST SP 800-22 battery (100 sequences × 10^6 bits, fresh key and nonce per sequence, no failures, §4.11), which rules out gross statistical biases but says nothing about differential, linear, algebraic, or key-recovery attacks. The related-key analysis is heuristic. No impossible-differential, boomerang, or algebraic analysis has been done. The cipher has not been externally audited.

Do not use MYSX to protect real data. For production: **AES-256-GCM** or **XChaCha20-Poly1305**.

11. Reproducibility

File	Purpose
`SPEC.md`	Formal specification (identical content to Section 2)

File	Purpose
`PAPER.md`	This document
`KAT.json`	Test vectors (block, AEAD, KDF)
`generate_kat.py`	KAT generator
`mysx.prod.js`	Implementation (public API)
`mysx.test.js`	Functional tests
`cryptanalysis/models/linear/mysx_linear_model.py`	Exact linear model of F + round; `selftest` / `report` commands
`cryptanalysis/models/linear/LINEAR_MODEL.md`	Detailed documentation of the exact linear model
`cryptanalysis/models/linear/linear_model_data.json`	All data in Section 4 (except §4.10) as JSON
`misc/mysx_linear_beam.py`	Beam search over the exact linear model, all offsets, up to 128 rounds, forward and backward (§4.10); `selftest`, `reproduce`, `run`, `merge`
`misc/beam_full.log`, `misc/beam_full.json`	Output of the forward §4.10 run (`run --beam 2000 --jobs 8`)
`misc/beam_bwd.log`, `misc/beam_bwd.json`	Output of the backward §4.10 run (`run --direction backward --beam 2000 --jobs 8`)
`misc/beam_pairs47.json`	Output of the pairs47 multi-bit run (merged; not an improvement)
`misc/verify_beam_trail.py`	Prefix-by-prefix check of a saved beam trail against the real cipher, both directions (§4.10.3)
`misc/hull_count.py`	Counts trails sharing the end masks of a saved beam trail (§4.10.3)
`misc/hull_bidir.py`	Budgeted exact linear-hull enumeration between two end masks (meet-in-the-middle; optional exact signs for random keys) (§4.10.5)
`misc/scan_endpairs.py`	Exact budgeted hull of the top-K end-mask pairs of a beam (§4.10.5)
`misc/test_signed_dp.py`	Validation of the signed hull of `hull_bidir.py` against `propagate()` (§4.10.5)
`misc/emp_check.py`	Empirical correlation of a saved beam trail against the real cipher, fixed key (§4.10.5)
`misc/beam_paths_bwd20.json`, `misc/beam_paths_fwd23.json`	Re-runs of the beam search (offset 20 backward, offset 23 forward, `--paths 15..22`), input of §4.10.5
`misc/scan_R17.log`	Output of the R=17 end-pair scan (§4.10.5)
`misc/HULL_R15_R17.md`	Summary and reproduction commands of §4.10.5

File	Purpose
misc/mysx_milp.py	Differential MILP (original, --r0 fixed at 0)
misc/mysx_milp2.py	Differential MILP with --r0 (§3.5)
misc/verify_diff_trail.py	Empirical verification of the 6-round deterministic trail at r0=15 (§3.5)
misc/verify_diff_13r.py	Empirical verification of the 13-round deterministic trail at r0=10 (§3.5)
misc/diff_free_chain_table.py	Differential free chain table by offset (§3.5)
misc/mysx_linear_milp.py	Previous approximate linear MILPs (historical, superseded by mysx_linear_model.py)
misc/mysx_division_property.py	Bit-based division-property MILP (v1; --xor-rule selects XOR or OR)
misc/mysx_division_property_v2.py	Bit-based division-property MILP (standard (1,1)-invalid XOR rule)
misc/verify_division.py	Empirical verification of integral distinguishers
misc/verify_division_double_toggle.py	Double-toggle verification for bits 5:6 and 7:9 (§4.9)
misc/keyschedule_test.js	Basic key schedule audit
misc/keyschedule_test_extended.js	Exhaustive audit (22 tests)
misc/check_trail.js	Empirical differential trail at r=4
misc/check_linear_trail2.js	Empirical check of the superseded linear model (negative result)
misc/validate_lineal_harness.js	Hand-computed linear harness check
misc/mini.js	Mini-MYSX (w=8, w=16, w=32)
misc/check_mini.js	Mini-MYSX validation vs mysx.prod.js
misc/check_mini_nondegenerate.js	Keyword/w compatibility check
misc/mini_linear_fast.js	Linear sampling at w=8, kw='MYSZ'
misc/verify_trail_w32.js	Per-trail verification at w=32 with MILP weight
misc/avalanche.js	Diffusion and avalanche ladder (r=1..24)
diffResults/	Raw logs of the differential offset sweep (§3.5)
dpResults/, dpResults_v2/	Raw logs of the integral offset sweep (§4.9)
misc/gen_nist_input.js	Generates the 100 × 10^6-bit CTR keystream files for NIST SP 800-22 (§4.11)
misc/nist_input/	Output of the above (100 files data000…data099, 125,000 bytes each)
misc/nist_input/all_data.bin	Concatenation of the 100 files (12,500,000 bytes) fed to ./assess 1000000

File	Purpose
NIST_SP_800-22_report.txt	STS 2.1.2 report (copy kept alongside the paper)

Fixed seed in tests (0xC0FFEE). MILPs are deterministic given the solver and gapRel. The mini-MYSX with w=32 reproduces mysx.prod.js exactly (F function and MYZ_PERM are bit-identical, verified in misc/check_mini.js). The exact linear model's selftest additionally verifies the Python port against KNOWN_KAT_DIGEST of mysx.prod.js, so the linear results are stated relative to the reference implementation.

Beam search (§4.10). The forward run is reproduced with python3 misc/mysx_linear_beam.py run --beam 2000 --jobs 8 --out beam_full.json; the backward run with the same command plus --direction backward --out beam_bwd.json. The combined table is reproduced with python3 misc/mysx_linear_beam.py merge beam_full.json beam_pairs47.json beam_bwd.json. The search is deterministic given the parameters (no randomness except a fixed seed inside the coset sampler for cells with rank(A_b) > 8). cmp confirms that two independent forward runs (beam_full.json and beam_full_1.json) are byte-identical. The hull checks of §4.10.3 use the saved R=14 trails with fixed keys (random.Random(seed) over 64 bytes, seed 1 and 2) and are not part of the tool; their scripts are misc/verify_beam_trail.py and misc/hull_count.py.

Exact hull (§4.10.5). From misc/: python3 test_signed_dp.py (validation against propagate()); python3 hull_bidir.py beam_paths_bwd20.json 20 --R 16 --wmax 14 --keys 2; python3 hull_bidir.py beam_paths_bwd20.json 20 --R 17 --wmax 18 --keys 2; python3 hull_bidir.py beam_paths_fwd23.json 23 --R 16 --wmax 10 --keys 2; python3 emp_check.py beam_paths_bwd20.json 20 16 28 1000 (about 12 min, one CPU); python3 scan_endpairs.py --R 17 --cells 20:backward,23:forward --topk 8. Keys are rand_key(1000), rand_key(1001), ...

Differential chains (§3.5). From misc/: python3 diff_free_chain_table.py (free chain table); python3 verify_diff_trail.py (6-round trail at r0=15); python3 verify_diff_13r.py (13-round trail at r0=10); python3 mysx_milp2.py --rounds 13 --input-bit 63 --r0 10 --trail (13-round trail).

Division-property results (§4.9). At r0=0, all three input patterns (single bit, word 0, words 0+4) were run with --r0 0. The (1,1)-invalid XOR rule and the OR-conservative rule were checked to agree on the tested patterns. Per-round balanced-bit counts are in the terminal logs of the runs; the runtimes were 0.4 s to 4.8 ks with SCIP. The R=10 single-bit dependency was verified against the real cipher (20,000 random samples per bit). The R=6 word-0 integral was verified against the real cipher at the last round (R=6, balanced bits confirmed; R=7 saturated). The R=5 words-0,4 integral was not verified empirically because its 2^{64}-text integral cannot be approximated by 20,000 random samples. At r0=15, the offset sweep used --r0 15 (all 31 offsets at the saturation rounds of r0=0). The R=7/R=6 bits at r0=15 were verified as consistent with bit-0 or double toggle; the full 2^{32}/2^{64} integrals were not verified.

Statistical tests (§4.11). From `misc/`: `MYSX_ENABLE_TEST=1 node gen_nist_input.js` regenerates the 100 files (the `TEST` API is enabled only by that environment variable and is not part of the public surface). Concatenate with `cat nist_input/data* > nist_input/all_data.bin`, then run `./assess 1000000` in STS 2.1.2 with input choice `0`, input file `.../nist_input/all_data.bin`, all tests, default parameters, 100 bitstreams, binary mode. Keys and nonces are drawn with `crypto.randomBytes` and are therefore not reproducible bit-for-bit; the *result* (no failures, all proportions inside the acceptance region) is stable across reruns, but a rerun produces different sequences. The report is deterministic given the input file.

12. Final warning

MYSX has not been externally audited. This paper is the author's self-analysis, not a certification. No result here replaces an independent cryptographic review.

Do not use MYSX to protect real data.